Jayanth Dasamantharao

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PROFESSIONAL EXPERIENCE

AI Engineer - Fedway, *EliteUS Software Solutions* — *Aug 2024 - Present | Piscataway, NJ*

- Developed an AI chatbot using **NLP, AWS Bedrock, LLMs, and Streamlit** for system instructions with Python. Leveraged **advanced language models** for seamless deployment and real-time communication.
- Integrated **vector databases, text-image matching, and LangChain** for dynamic interactions and enhanced response accuracy through **Retrieval-Augmented Generation (RAG)**.
- Tech Stack: Python, AWS Bedrock, Azure, OpenAI, LangChain, Chroma, Vector db, PyMuPDF, Pillow.

Data Science Intern, *Harvest Software Solutions, LLC* — *June 2023 - Sept 2023 | Jacksonville, Florida*

- Optimized real-time ad campaigns by deploying **H2O AutoML and XGBoost models**, achieving a 70% boost in predictive accuracy and ad spend efficiency.
- Leveraged **FastAPI** to deploy NLP models with Hugging Face Transformers for customer sentiment and topic analysis on a 5M+ row dataset, providing actionable insights on engagement using python.

Application Development Associate, *Accenture Solutions Pvt. Ltd.* — *June 2021 - Aug 2022 | Hyderabad, India*

Database Analyst, Regeneron Pharmaceuticals Inc. - Client Data Analysis & Warehousing

- Optimized **ETL** workflows and implemented **automated data pipelines in Python**, reducing support calls by 50% and enhancing data processing efficiency.
- Performed large-scale data analysis using **SQL** and **AWS Athena, Redshift** on large scale datasets; visualized insights and built interactive dashboards in **Tableau and Power BI**.
- Increased system reliability by 25% through **DDL optimization, seamless data integration, and performance tuning**, ensuring stable **data warehousing** and reporting operations using **SQL Workbench**.

Research Lab, *Andhra University* — *January 2021 - June 2021 | Visakhapatnam, India*

Self Driving Cars Using Image Processing and Deep Learning

- Engineered **Convolutional Neural Networks** (CNNs) for autonomous lane detection and object recognition using python.
- Leveraged **YOLO and Faster R-CNN** architectures to optimize real-time detection and classification accuracy.
- Enhanced model generalization through **image pre-processing** techniques like **Gaussian blurring and color space conversion**, and integrated **multi-sensor data** from **radar, lidar, and cameras** to improve navigational precision.
- Developed a real-time simulation environment using **Unity3D and a Flask backend** for dynamic parameter monitoring and system validation, enabling high-fidelity testing of autonomous driving functionalities.

TECHNICAL SKILLS

Programming: Python, R, Scala, MATLAB. **Libraries:** Pytorch, Scikit-learn, Tensorflow, SparkML, Tableau, PowerBI.
Machine Learning: Ensemble Methods, Classification models, Gridsearching, Clustering, Regression Analysis, LightGBM.
Statistical Analysis: Causal Inference, Hypothesis testing, Time-series forecasting, stochastic process, Linear Regression.
Deep Learning: Neural Networks, Computer Vision, AI, NLP, BERT, GANs, AWS Bedrock, Azure, OpenAI models.
Language Models: OpenAI, Azure, AWS Bedrock, Hugging Face Transformers, LLaMA, prompt tuning, RAG, Langchain.
Data Engineering: SQL, PostgreSQL, Tableau prep, Kafka, Apache Spark, Data Warehousing, Hadoop, GCP, Docker.

EDUCATION

Rutgers University, *Masters in Data Science* — *Sept 2022 - May 2024 | New Brunswick, NJ*

Relevant Coursework: Statistics: Statistical Modeling and Computing, Stat Learning; Neural Networks, Machine Learning, Data Structures and Algorithms(Python), Data Wrangling (R), NLP, Data Mining. **Projects Overview:** Portfolio.

Andhra University, *B.Tech in Electrical and Electronics Engineering* — *2017 - 2021 | Visakhapatnam, India*

RESEARCH PROJECTS

Bayesian Classification - [Bayesian Networks, PyMC3, Informative Prior Distributions] Developed Bayesian logistic regression models using PyMC3 to predict diabetes status, incorporating informative priors. Evaluated performance with recall and f1-score metrics, comparing outcomes between uniform and normal distribution priors.

Language Identification - [Naive Bayes, Logistic Regression, BERT, GloVe embeddings] Worked on identifying languages using Naive Bayes, Logistic Regression, and DistilBERT, incorporating n-grams analysis achieving 96% accuracy.

Auto Insurance Fraud Detection - [Python, Data Visualization, ML models - Random Forests, KNN, SVM, Decision Trees, XGBoost, Regression] Executed diverse ML strategies and time series analysis for auto insurance fraud, prioritizing reduced false negatives. Chose K-Nearest Neighbors (KNN) with an impressive 97% recall.

Thank you for considering my application. I'm available to start working immediately.